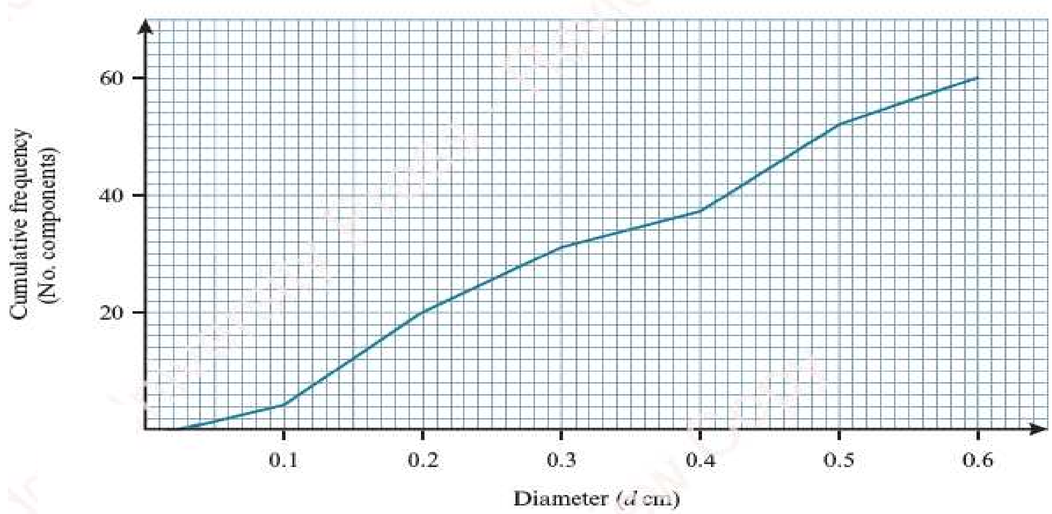
United International University
School of Science and Engineering
Department of Computer Science and Engineering

Mid Term Examination || Trimester: Fall – 2025
Course: Math -2205 (Probability & Statistics)
Total marks – 30 || Duration – 1 hour 30 minutes

[Note *that the number of marks is given in brackets* [ ] *at the end of each question or part question. You are requested to answer all the questions in order*.]

Q1 **(a)** The diameters, *d* cm of 60 cylindrical electric components, are represented in the following cumulative frequency graph



(i) Find the median and inter-quartile range of the diameter of the cylindrical electric components. [3]

(ii) Showing necessary working, state the nature of the distribution [2]

(b) Estimate the number of components that have a diameter less than 0.15 cm [1]

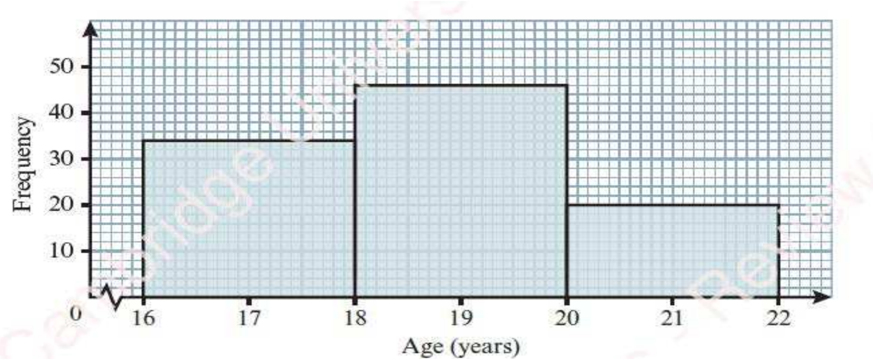
(c) Estimate the value of *K*, given that 20% of the components have diameter of *K* cm or more [2]

Q2 **(a)** The following data set represents the scores obtained by 12 students in a mathematics test:

45, 48, 50, 52, 53, 55, 56, 58, 60, 62, 95, 97

Using an appropriate statistical method, to determine whether there exists any **possible outlier** in the data set. [3]

(b) The ages, in years of some children are represented in the following histogram



Construct a frequency distribution based on the above histogram and hence find the mean age of the children. [4]

P.T.O.

Q3 **(a)** Assume that the data given in the following table represents the GPA distribution of 40 students of UIU.

GPA	0 – 1	1 – 2	2 – 3	3 – 4
Number of students	6	f	18	9

(a) Find the value of f. **[1]**

(b) Estimate the mode of the distribution. **[3]**

(c) Calculate the percentage of students who will not receive a penalty, given that students with a GPA less than 1.00 will receive a penalty. **[1]**

(b) Two sections of a university course, Section A and Section B, recorded the following statistics for the number of hours students spend studying per week:

Section	**Mean study hours ($\bar{x}$)**	**Standard deviation (σ)**
A	25 hours	5 hours
B	30 hours	9 hours

Calculate the coefficient of variation (CV) for both sections and hence identify which section shows greater consistency in study hours. **[3]**

Q4 **(a)** What kind of correlation exists between relations: **[2]**

(i) daily temperature and ice-cream sales

(ii) age of a car and its resale value

(b) The ranks of 6 delivery agents based on delivery speed and customer rating are given: **[3]**

Agent	A	B	C	D	E	F
Speed Rank	1	2	3	4	5	6
Rating Rank	2	1	4	3	6	5

Calculate Spearman's rank correlation coefficient and interpret the result.

(c) A logistics company studies the relationship between distance traveled ($\boldsymbol{x}$, in km) and fuel consumption ($\boldsymbol{y}$, in liters) for its delivery vehicles. Using past data, the least squares regression line of fuel consumption on distance is obtained as: **[2]**

$$\boldsymbol{y = 2.5 + 0.08\,x}$$

(i) Interpret the **slope** of the regression line in the context of this problem

(ii) Estimate the fuel consumption for a trip of 50 km using the given regression line

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Formulae:	For the regression line: $y = a + bx$	$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n\sum x^2 - (\sum x)^2}$ and $b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$
	Coefficient of correlation	$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}}$
	Spearman's rank correlation coefficient:	$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$